

Tunable Consistency and Client-Centric Guarantees in MongoDB

Project 1 · Complete Experimental Report

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Experimental platform: Google Cloud Compute Engine (Singapore)

Main finding: causal sessions with majority reads and writes preserved all four guarantees for successful operations; weak reads exposed all three read-related anomalies under controlled replication lag.

Repository: github.com/Panyuxiang0830/DSA5208_project1

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Abstract

This project deploys a three-member MongoDB 8.0.29 replica set with Docker Compose on one Google Cloud virtual machine and studies how read concern, write concern, read preference, and causally consistent sessions affect four client-centric guarantees: read-your-writes, monotonic reads, monotonic writes, and writes-follow-reads. The experiment suite covers normal operation, a stopped secondary, a stopped primary, a primary-side network partition, continuous requests during elections, and controlled replication lag. Each formal baseline and stable-fault matrix uses 500 sequences and three random seeds for every configuration-model pair; transition tests run for 35 seconds; the lag test freezes apply processing on one readable secondary. C1, C2, and C4 showed no consistency violations in successful checks. Weak configuration C3 produced a 76.07% read-your-writes violation rate in normal operation and 64.60%-69.93% under stable faults. Election windows caused transient operation errors but no new successful-read anomaly in the strong configurations. Under S4, C3 violation rates rose to 83.33% for read-your-writes, 49.80% for monotonic reads, and 50.00% for writes-follow-reads, while monotonic writes remained intact. These observations support MongoDB's documented causal-consistency behavior, while the absence of observed violations remains experimental evidence rather than a proof over all histories.

Keywords: MongoDB; replica sets; tunable consistency; causal consistency; fault injection; replication lag

1. Objectives and Research Questions

The assignment requires a replicated database with tunable consistency, predictions about client-observed guarantees, and experiments under normal operation, node failures, and a network partition. MongoDB was selected because its driver exposes sessions, read concern, write concern, and read preference, allowing abstract consistency models to be mapped to reproducible client configurations.

- Which configurations preserve read-your-writes (RYW), monotonic reads (MR), monotonic writes (MW), and writes-follow-reads (WFR)?
- How do stable degraded states and election transition windows differ in their effects on consistency and availability?
- Can controlled replication lag amplify occasional stale reads into stable, explainable anomalies?

2. Background: Four Client-Centric Guarantees

The four guarantees are complementary session guarantees within causal consistency, not a simple one-dimensional strength ranking. A system may satisfy one while violating another; causal consistency is commonly characterized by all four together. The table gives the executable checks used here.

Model	Application expectation	Operational check
RYW	After writing, the client reads that version or newer.	<code>observed_version < acknowledged_version</code>
MR	Successive reads by one client never go backwards.	<code>later_version < earlier_version</code>
MW	Writes from one client take effect in issue order.	<code>acknowledged history/final version out of order</code>
WFR	A dependent write is ordered after the version it read.	<code>parent missing/stale predicate/dependent write missing</code>

3. Database and Deployment Architecture

3.1 Software and Cloud Resources

Item	Configuration
Cloud	Google Cloud Compute Engine; project nifty-pursuit-505911-n0
VM	dsa5208-mongodb; asia-southeast1-b (Singapore); e2-standard-2; 2 vCPU; 8 GB RAM
OS	Ubuntu 24.04.4 LTS; Linux 6.17.0-1022-gcp
Storage	30 GB balanced persistent disk; MongoDB named volumes
Containers	Docker Engine 29.7.2; Docker Compose v5.5.0
Database	MongoDB Community 8.0.29; three data-bearing voting members; no arbiter
Client	Python 3.12-slim; PyMongo 4.17.0
Limits	1536 MB RAM per mongod container; 0.5 GB WiredTiger cache

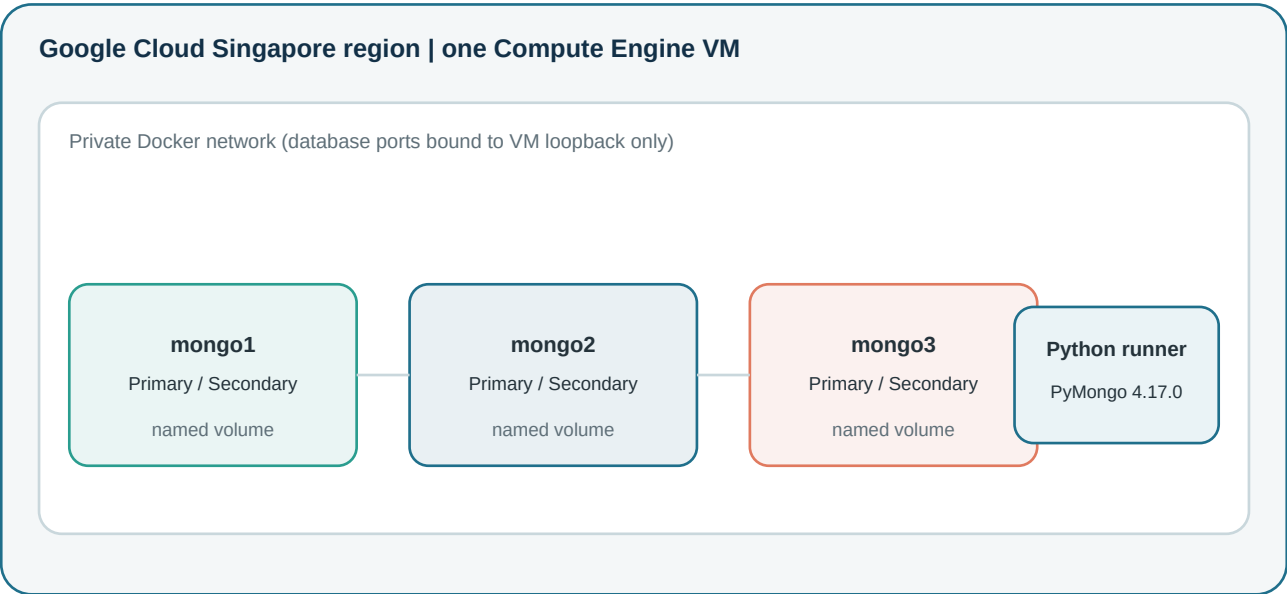


Figure 1. Experimental deployment. The three MongoDB members use separate containers, processes, and persistent volumes, but share one VM.

This design provides a logically replicated deployment and deterministic container-stop and container-network fault injection. Database ports are bound only to VM loopback and are not exposed directly to the public network. Its main trade-off is shared host, disk, and underlying network resources, so it does not model physical fault independence.

3.2 Installation and Initialization

```
git clone https://github.com/Panyuxiang0830/DSA5208_project1.git
cd DSA5208_project1
./scripts/bootstrap-ubuntu.sh
cp .env.example .env
docker compose pull
docker compose up -d --wait mongo1 mongo2 mongo3
docker compose run --rm mongo-init
./scripts/cluster_status.sh
docker compose build runner
docker compose run --rm --no-deps runner
```

The initialization script creates three voting, data-bearing replica-set members. Under normal operation one member is Primary and two are Secondary; MongoDB's replica-set protocol controls elections. Removing or restarting containers does not remove named volumes, so database files survive normal recovery procedures.

4. Consistency Configurations and Predictions

MongoDB write concern controls when replication acknowledgment is sufficient, read concern controls the visibility level that reads may observe, read preference routes reads to primary or secondary members, and causal sessions carry causal time through the driver. MongoDB documents that causally consistent sessions combined with majority read and write concerns provide RYW, MR, MW, and WFR across members.

Config	Session	Write concern	Read concern	Read preference	Prediction
C1	Causal	majority	majority	primary	All four hold
C2	Causal	majority	majority	secondary	All four hold; may wait for replica
C3	Non-causal	w: 1	local	secondaryPreferred / Secondary	RYW/MR/WFR may fail; MW usually holds
C4	Non-causal	majority	majority	primary	Expected to hold here; not equivalent to general causal session

C3 is intentionally weak: w:1 requires acknowledgment from one mongod, local reads need not be majority committed, and a secondary may lag the primary. C4 separates the effect of majority/primary access from causal sessions. It is strong for this single-client, primary-directed workload but should not be generalized to every cross-client causal history.

5. Experimental Design

5.1 Workloads and Measurement

Each test document carries a monotonically increasing version, writer/run identifiers, and dependency fields. The runner records operation start/end times, results, errors, serving member, and model verdicts, then emits raw JSONL, summary JSON, and Markdown. Every formal baseline/stable-fault run executes 500 sequences per configuration-model pair for seeds 20260830, 20260831, and 20260832, yielding 1,500 principal checks. Database operation errors are counted separately from consistency violations: an error is an unavailable request, while a violation is evaluated only on a successful, decidable history.

Scenario	Fault or control	Purpose
S0	No fault	Normal-operation baseline
S1	Stop one Secondary	Stable behavior with a live majority
S2	Stop old Primary; wait for election	Post-election two-member topology
S3	Disconnect old Primary from majority and client	Stable state after a primary-side partition
T1	Continuous RYW during Primary stop	Consistency and availability during election
T2	Continuous RYW during Primary partition	Partition-triggered election window
S4	Freeze apply on one readable Secondary	Amplify explainable stale reads

5.2 Fault Injection and Recovery

S1 and S2 stop the container holding the selected role. S3 uses docker network disconnect to isolate the pre-fault Primary from the replica-set network; the remaining two members retain a majority and elect a new Primary. T1/T2 start concurrent loops for all four configurations before fault injection and label requests by operation start time as pre, election, or post. S4 enables `enableTestCommands=1` only in a test-specific Compose override and runs `rsSyncApplyStop` on one Secondary; afterward, apply processing is resumed, catch-up is verified, and normal containers are recreated. Every scenario ends with `restore_cluster.sh` and `cluster_status.sh` to confirm a healthy three-member set.

Rationale: S0-S3 isolate stable-state consistency, T1/T2 expose the transient unavailability hidden by waiting for election completion, and S4 provides a controlled causal mechanism showing that C3 anomalies arise from replica lag rather than test noise.

6. Results

6.1 S0-S3: Stable-State Matrix

S0 completed 60,012 database operations and 18,012 consistency checks with 0 errors. S1, S2, and S3 completed 18,012, 18,012, and 18,012 checks respectively, with 0, 0, and 0 database-operation errors. C1, C2, and C4 recorded 0 violations combined across every model and scenario; the only stable C3 anomaly was RYW.

Scenario	C3 RYW	Rate	Other C3 models	C1/C2/C4
S0	1,141 / 1,500	76.07%	MR 109/1,500 (7.27%); MW 0; WFR 0	0
S1	1,035 / 1,500	69.00%	MR 0; MW 0; WFR 0	0
S2	969 / 1,500	64.60%	MR 0; MW 0; WFR 0	0
S3	1,049 / 1,500	69.93%	MR 0; MW 0; WFR 0	0

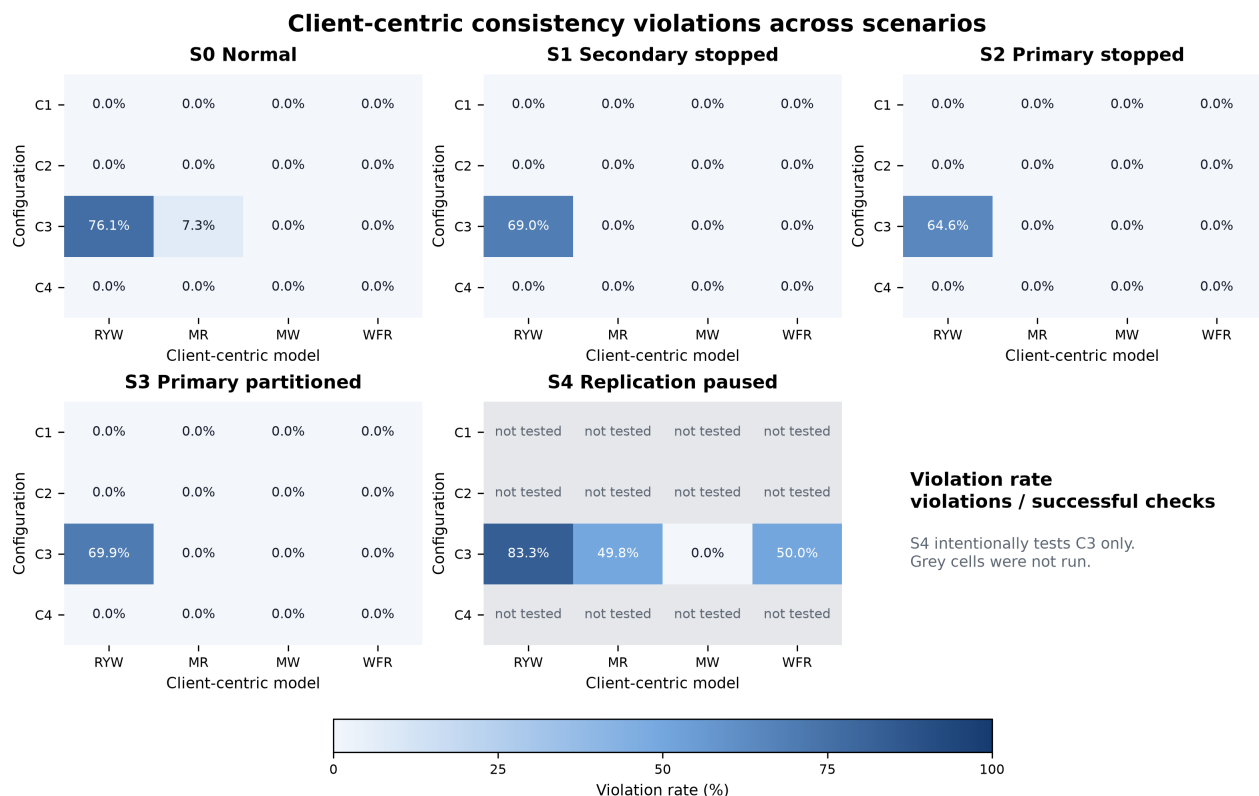


Figure 2. Consistency-violation matrix for S0-S4. Color encodes the violation rate among 1,500 principal checks.

The slightly lower C3 RYW rates in S1-S3 do not mean faults strengthened C3. The number of available replicas, the selected Secondary, and runtime scheduling changed the chance of choosing a stale member. These rates are empirical frequencies in this deployment, not protocol guarantees.

6.2 Latency

Config	S0 p50 / p95 (ms)	S1 p50 / p95	S2 p50 / p95	S3 p50 / p95
C1	3.19 / 9.24	2.09 / 5.84	2.23 / 7.09	2.05 / 6.07
C2	3.43 / 9.11	2.21 / 6.62	2.84 / 9.73	2.15 / 7.04
C3	1.48 / 5.00	1.01 / 3.35	1.07 / 3.14	0.88 / 2.85
C4	3.22 / 9.13	2.25 / 6.79	2.24 / 6.65	2.17 / 6.45

The table reports end-to-end latency for one RYW write-read pair. C3 is fastest, consistent with w:1/local avoiding majority waits, but this accompanies weaker visibility. C2's baseline WFR p95 was 42.27 ms versus 19.37 ms for C1, consistent with a causal secondary read sometimes waiting to satisfy afterClusterTime. Lower medians in some fault

scenarios reflect container, cache, and runtime variation and should not be interpreted as fault-induced optimization.

6.3 T1/T2: Election Transition Windows

Scenario	Checks	Violations	Op. errors	Mean election	Strong successful ops
T1	9,635	2,420 (all C3)	34 (all election)	11,334 ms	0 violations
T2	10,974	2,532 (all C3)	60 (all election)	15,041 ms	0 violations

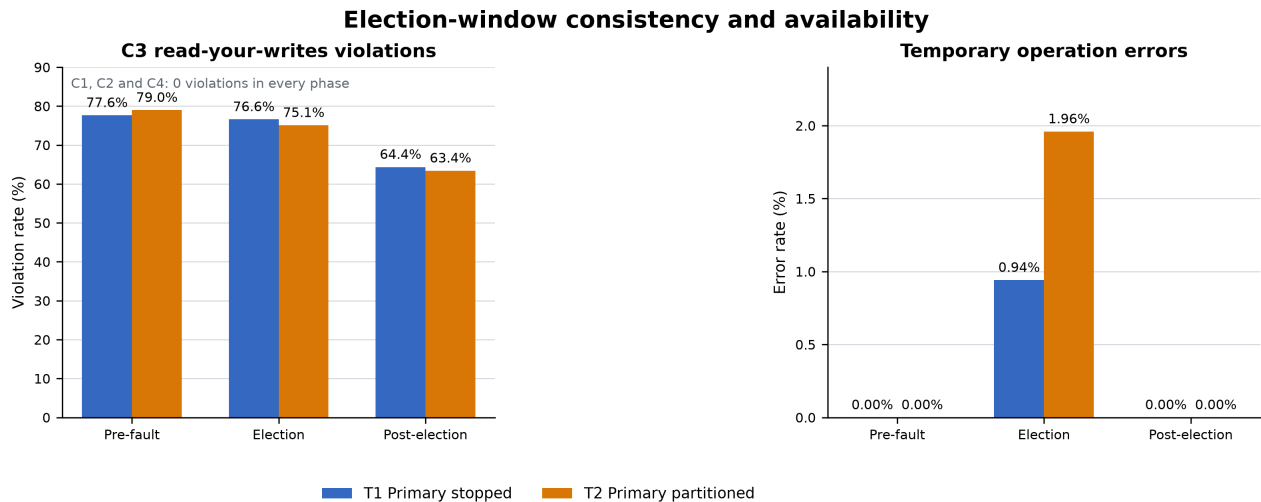


Figure 3. Successful checks, violations, and operation errors by configuration in T1/T2.

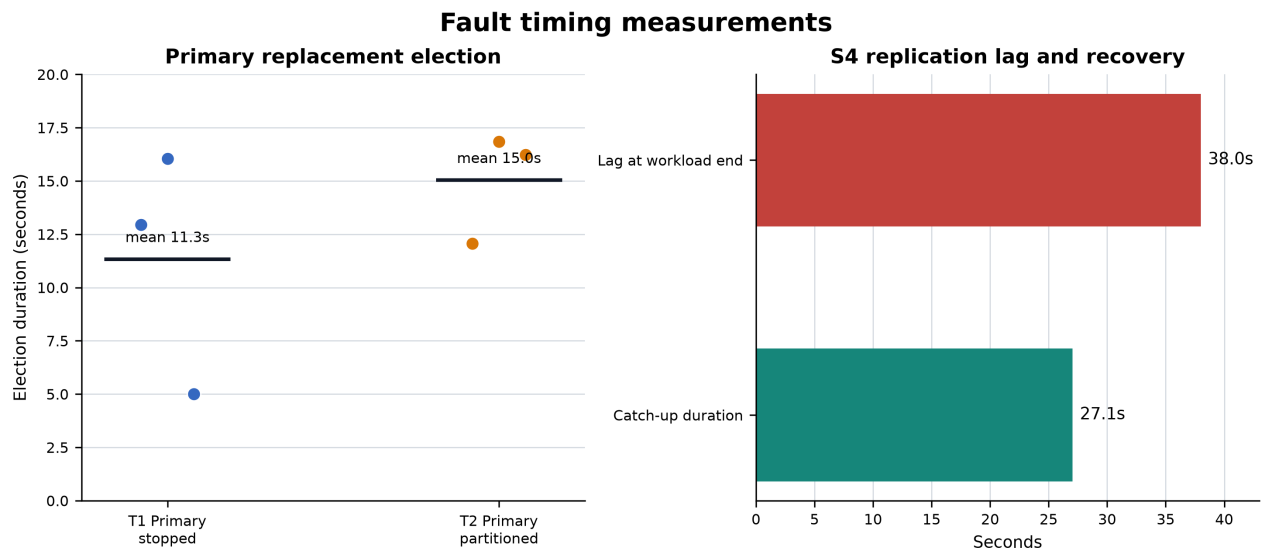


Figure 4. Election timing for S2, S3, T1, and T2. T1/T2 show all three seeds.

Scenario	pre	election	post
T1 C3	771/993 = 77.64%	537/701 = 76.60%	1,112/1,728 = 64.35%
T2 C3	849/1,075 = 78.98%	413/550 = 75.09%	1,270/2,003 = 63.40%

The transition tests give two complementary conclusions. First, successful RYW pairs under C1, C2, and C4 recorded 0 violations combined. Second, 34 and 60 operation errors occurred during elections, showing a temporary availability cost when no writable Primary could be selected immediately. C3 was already weak before each fault, with similar pre and election violation rates; the election did not create a new C3 consistency defect, but mainly added transient failures. The per-seed election-window error count varied noticeably (T1: 3-27; T2: 16-24), which on a single shared VM more likely reflects scheduling noise than a real difference in protocol behaviour, so it is reported here rather than treated as a stable quantitative result.

6.4 S4: Controlled Replication Lag

Model	Violations / checks	Rate	Interpretation
RYW	1,250 / 1,500	83.33%	Post-write read hits the frozen Secondary.
MR	747 / 1,500	49.80%	Alternation between current and frozen secondaries.
MW	0 / 3 validations	0.00%	Primary accepts ordered writes; final state remains ordered.
WFR	750 / 1,500	50.00%	Dependent write is based on a stale parent read.

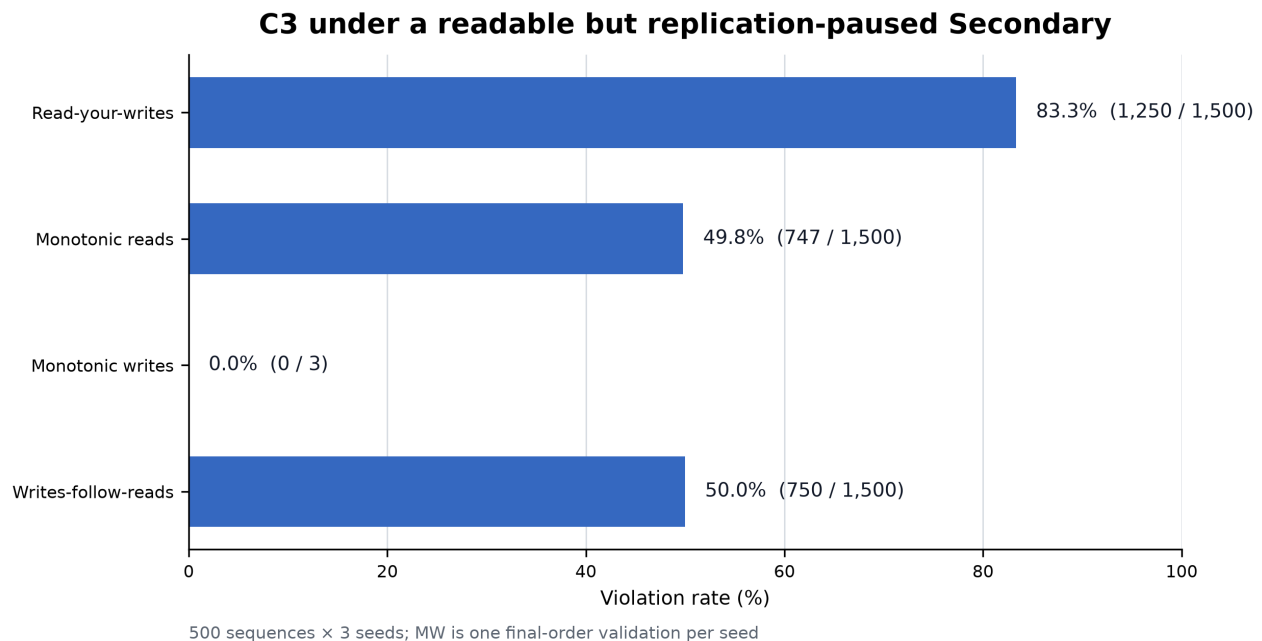


Figure 5. Per-model violation rates for weak configuration C3 under S4.

S4 completed 4,503 checks with 0 database-operation errors and 2,747 violations (61.00%). The frozen member was 38 seconds behind at test end. After apply resumed, three consecutive status checks each found lag within one second; the full recovery took 27,051 ms. MR and WFR near 50% match alternating selection between one frozen and one current Secondary, providing strong mechanistic evidence.

6.5 Controlled Concern Matrix and Session Ablation

C1-C4 changed session, write concern, read concern, and read preference all at once, so C3's weak results cannot be pinned on any single setting. C5-C8 add four configurations with read preference fixed at secondary: C2, C5, C6, and C7 form a 2x2 write/read-concern matrix under a causal session, and C8 repeats C2's concern settings with the session turned off. The first four isolate what write concern and read concern each contribute once session and read preference are fixed; C8 isolates what the session itself contributes once concern and read preference are fixed.

Config	Session	Write concern	Read concern	Prediction
C2	Causal	majority	majority	All four durably guaranteed
C5	Causal	w: 1	majority	MR, WFR durably guaranteed; RYW, MW not
C6	Causal	majority	local	MW guaranteed; RYW, MR, WFR not
C7	Causal	w: 1	local	None of the four guaranteed
C8	Non-causal	majority	majority	No general session guarantee for any; MW may still hold in this workload

Across S0-S3, C5, C6, and C7 recorded zero violations on every model (not tabulated separately): with a healthy topology and a causal session, even weak write/read concern avoided observable violations in this workload. That only partly matches the prediction. C7 (w:1, local) was predicted to guarantee none of the four models, but no violation appeared without a fault or artificial lag, so a zero-violation result here is workload-dependent evidence, not

proof that the session is strong. C8 is the exception: under fault-free S0 it already showed a 14.73% read-your-writes violation rate (221/1,500), dropping to 0.07% under the stable S1 fault (1/1,500). Majority write concern plus majority read concern, without a causal session, does not guarantee read-your-writes: majority only guarantees durability, not that the next Secondary the client happens to read from has already applied that write.

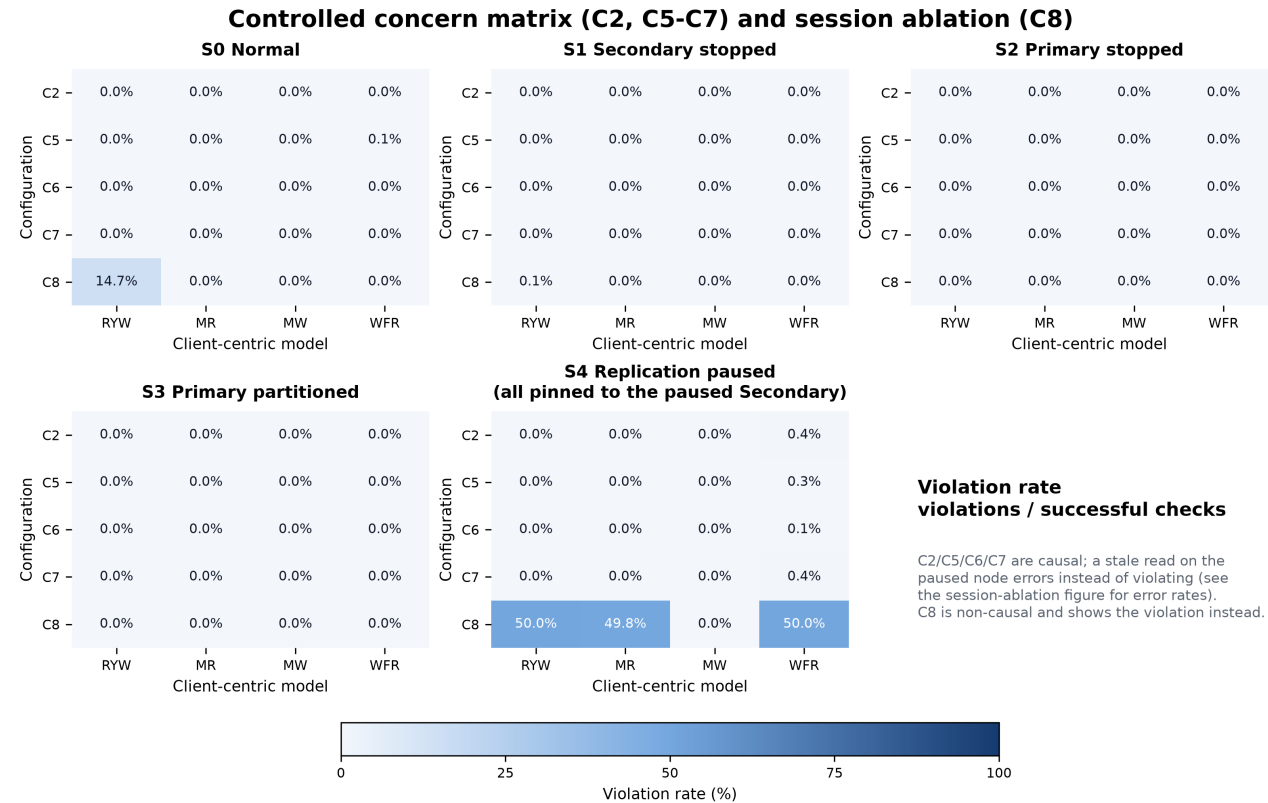


Figure 6. Violation-rate matrix for C2, C5-C8 across S0-S4; under S4 every configuration reads the same Secondary with paused oplog application.

Config	RYW rate/err	MR	MW	WFR
C2	0.00% / err 33%	0.00% / err 33%	0.00% / err 0%	0.41% / err 51%
C5	0.00% / err 34%	0.00% / err 33%	0.00% / err 0%	0.27% / err 50%
C6	0.00% / err 33%	0.00% / err 33%	0.00% / err 0%	0.13% / err 49%
C7	0.00% / err 34%	0.00% / err 35%	0.00% / err 0%	0.41% / err 51%
C8	50.00% / err 0%	49.80% / err 0%	0.00% / err 0%	50.00% / err 0%

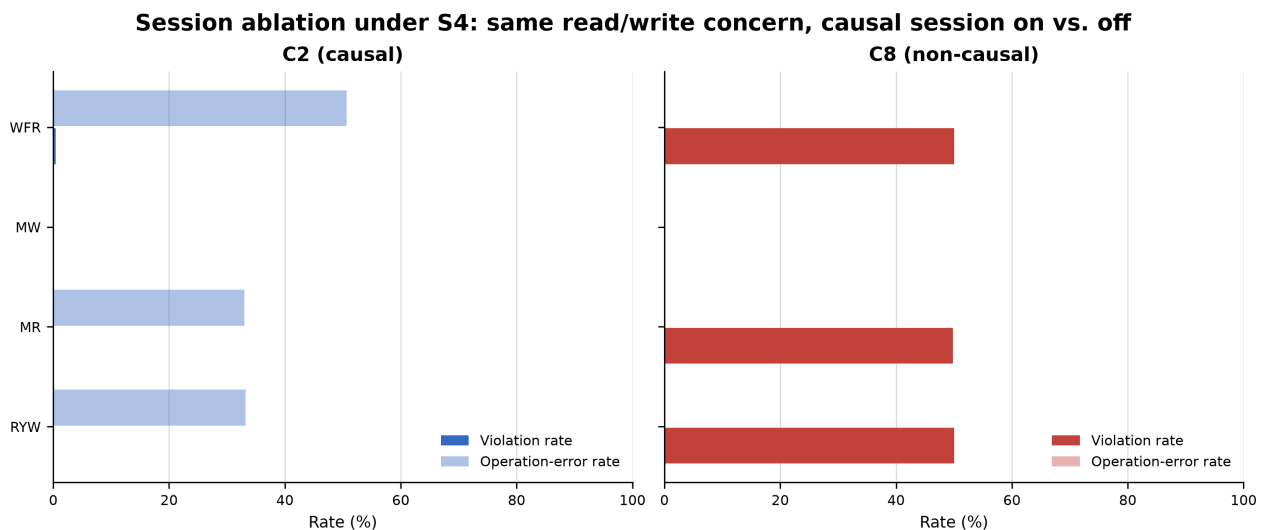


Figure 7. C2 versus C8 under S4: identical write/read concern and read preference, differing only in the causal session.

S4 holds read preference at Secondary and routes all six configurations to the same node paused with `rsSyncApplyStop`, so any difference in the matrix comes only from session and concern settings. With a causal session on (C2, C5, C6, C7), RYW/MR/WFR violation rates are all 0%, but operation-error rates run from roughly 33% to 51%: a causal read that cannot satisfy `afterClusterTime` waits and times out instead of returning a stale version. Write/read concern still matters among these four configs, but mainly through error-rate magnitude and the latencies measured in 6.1-6.4, not through whether the four guarantees hold. C2 and C8 make the session's effect explicit: with identical write/read concern and read preference and only the session differing, C8's WFR violation rate is 50.00%, while C2's is 0.41% and its operation-error rate is 51% instead. The causal session does not stop the read from going stale; it stops the stale read from succeeding.

Violation counts in the T1/T2 transition windows (3 seeds each, C5-C8 continuously sending RYW write-then-read pairs) were: C5 T1 0, T2 0; C6 T1 0, T2 0; C7 T1 0, T2 0; C8 T1 18, T2 9. C5-C7 (causal) match the C1/C2/C4 pattern of no violations in successful histories. C8 (non-causal) showed a handful of scattered violations, on the same order as its S0/S1 normal-operation results; the election transition did not introduce a new failure mode for C8, it just continued the weakness that was already there.

7. Predictions Versus Observations

Config	Prediction	Observation	Assessment
C1	All four hold	Zero violations in S0-S3 and successful T1/T2 RYW	Agrees
C2	Cross-member causal session holds	Zero violations on Secondary reads; some higher tail latency	Agrees
C3	RYW/MR/WFR may fail; MW usually holds	S0 exposed RYW/MR; S4 exposed RYW/MR/WFR; MW zero	Agrees; S4 completes mechanism
C4	Holds here, not a general causal guarantee	Zero violations in this workload	Supported within scope

The key distinction is safety versus availability. The strong configurations did not convert election-time failures into consistency anomalies: unsafe operations failed or retried, while successful histories continued to satisfy RYW. C3 instead offered low-latency, highly available reads that could successfully return stale versions.

8. Limitations and Threats to Validity

- All members share one VM, hardware, disk, and underlying network; container isolation is not equivalent to three physical or virtual hosts.
- S1-S3 run the full matrix after election completion. T1/T2 cover only RYW continuously during the transition; the other three models were not stressed through elections.
- S3 isolates the old Primary from both peers and the client. It does not create split-client histories with clients on both partition sides.
- S4 uses `enableTestCommands` and `rsSyncApplyStop` as an extreme diagnostic failpoint, not as a realistic latency distribution.
- Only three seeds are used. Workload shape, driver selection, and local scheduling influence empirical rates. Zero observed violations is not a mathematical proof.
- Phases are assigned by operation start time; an operation crossing an election boundary still belongs to one phase.
- MW performs 500 sequential writes per configuration and seed, but the summary denominator counts one final history validation per seed; therefore the table reports three validations.
- Cloud billing has reporting lag. The VM was stopped after experimentation. The console showed USD 0.14 gross month cost, USD 0.14 credits, zero net cost, and USD 49.64 of the USD 50 credit remaining (about 99%).
- The formal C5-C8 data was collected on a local macOS workstation (a virtualized Linux VM inside Docker Desktop), not the GCP VM used for C1-C4. Container networking and disk characteristics differ between the two environments, so comparisons within 6.5 (C2 and C5-C8 against each other) are reliable, but comparing C5-C8's absolute numbers directly against C1-C4's in 6.1-6.4 across environments should be done cautiously, especially for latency-sensitive checks such as RYW.
- `max_time_ms` for causal configurations under S4 is set to 300 ms to bound how long the client waits before declaring failure, keeping the 500x3 formal run tractable. It does not measure how long causal consistency actually needs to catch up; the run only records whether an operation timed out.

9. Reproducibility

9.1 Main Commands

```
# Baseline S0
docker compose run --rm --no-deps runner python -m experiments.run_baseline \
--iterations 500 --seeds 20260830,20260831,20260832 --label formal

# Stable fault scenarios (replace the scenario name for S2/S3)
./scripts/run_fault_scenario.sh S1-secondary-failure 500 \
20260830,20260831,20260832 formal

# Transition windows; repeat for seeds 20260830-20260832
./scripts/run_transition_scenario.sh T1-primary-stop-transition 20260830 formal 35 3
./scripts/run_transition_scenario.sh T2-primary-partition-transition 20260830 formal 35 3

# Controlled replication lag (default configs: C2,C3,C5,C6,C7,C8)
./scripts/run_replication_lag_scenario.sh 500 \
20260830,20260831,20260832 formal

# C5-C8 controlled concern-matrix extension (6.5): same commands, restricted
# to the new configs via --configs (run_fault_scenario.sh and
# run_replication_lag_scenario.sh take it as a trailing positional argument;
# transition_window.py, invoked by run_transition_scenario.sh, takes it as
# a flag and otherwise defaults to every config in experiments.common.CONFIGS).
```

```

docker compose run --rm --no-deps runner python -m experiments.run_baseline \
--configs C5,C6,C7,C8 --iterations 500 \
--seeds 20260830,20260831,20260832 --label formal

# Recovery and report figures
./scripts/restore_cluster.sh
./scripts/cluster_status.sh
python3 -m analysis.generate_report_figures

```

The repository contains Compose configuration, initialization and fault scripts, Python workloads, formal summary JSON, result narratives, and chart-generation code. Raw JSONL can be regenerated with the commands above. To avoid unintended compute charges, stop the VM in Google Cloud after experiments and verify TERMINATED status; persistent disk may continue to incur storage charges.

9.2 Formal Run Identifiers

Scenario	Selected formal run ID / suffix
S0	baseline-s0-formal-20260830T154711Z-3c931c32
S1	s1-secondary-failure-formal-20260830T181846Z-8e657452
S2	s2-primary-failure-formal-20260830T182320Z-f604f5eb
S3	s3-primary-partition-formal-rerun-20260830T183455Z-59f3d378
S4	s4-secondary-replication-lag-formal-20260831T081523Z-85b1e3b3
T1	6cad906c; 867b572c; ae94f015
T2	0609280f; b9f2cd1b; ceb526e5
S0 (C5-C8)	s0-normal-ext-formal-20260915T150614Z-0036db8b
S1 (C5-C8)	s1-secondary-failure-ext-formal-20260915T150711Z-086fb8b5
S2 (C5-C8)	s2-primary-failure-ext-formal-20260915T151407Z-729c803d
S3 (C5-C8)	s3-primary-partition-ext-formal-20260915T152115Z-3652c475
S4 (C2,C3,C5-C8)	s4-secondary-replication-lag-ext-formal-20260915T153446Z-9a202a18
T1 (C5-C8)	bbcaf6f4; aaa87588; bba0e6f9
T2 (C5-C8)	71fcb5ca; 95b1ed82; 91c35c32

10. Conclusion

The project completes a full cycle from theoretical prediction through replica-set deployment, tunable configurations, four executable model workloads, fault injection, and mechanistic validation. Strong MongoDB session configurations preserved all four client-centric guarantees in every successful history observed. Weak secondary reads returned faster but could return older versions during normal replication, around elections, and under controlled lag. The most direct transition effect of node failure and partition was temporary unavailability rather than degradation of successful operations under strong configurations. Finally, S4 amplified RYW, MR, and WFR failures while preserving MW, demonstrating that these session guarantees are complementary and independently observable.

References

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All online documentation was last accessed on 31 August 2026.

AI Usage Statement

OpenAI ChatGPT/Codex was used to help explain course concepts, plan experiments, draft and review scripts, organize results, generate charts, and edit the report. Cloud-resource creation, command execution, fault injection, result-file inspection, source verification, and final conclusions were supervised and checked by the project member. Experimental values in this report come from the stored run outputs and were not fabricated by AI. The project member remains responsible for the submitted content, interpretation, and any errors.